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DD FORM 672-1

PROCUREMENT SPECIFICATION FF-S-86

P.A. Navy
Other Cust
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SUPERSEDING

TITLE

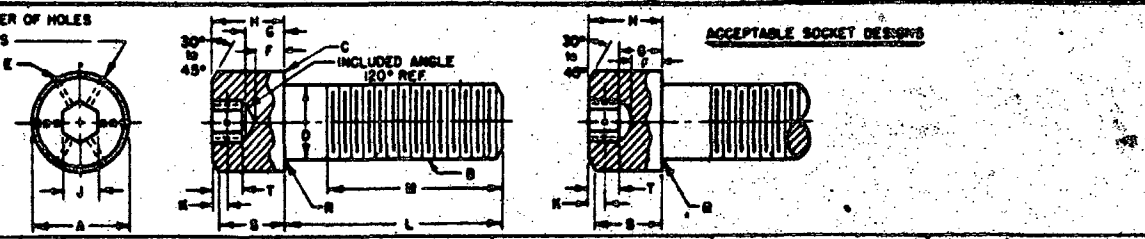
SCREW, CAP—SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL, UNCOATED (UNF-3A)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

MILITARY STANDARD

MS 24675

SHEET 1 OF 3




| NOMINAL SIZE | | #6 (.138) | #8 (.164) | #10 (.190) | 1/4 | 5/16 |
|---|---|---|---|---|---|---|
| B- THREADS PER INCH | | 40UNF-3A | 36UNF-3A | 32UNF-3A | 28UNF-3A | 24UNF-3A |
| D- BODY DIAMETER | MAX | .1380 | .1640 | .1900 | .2500 | .3125 |
| | MIN | .1329 | .1585 | .1840 | .2435 | .3053 |
| A- HEAD DIAMETER | MAX | .226 | .270 | .312 | .375 | .468 |
| | MIN | .218 | .262 | .303 | .365 | .457 |
| H- HEAD HEIGHT | MAX | .138 | .164 | .190 | .250 | .312 |
| | MIN | .134 | .159 | .185 | .244 | .306 |
| F- HOLE DEPTH | MIN | .013 | .011 | .016 | .026 | .027 |
| S- HEAD SIDE HEIGHT | MIN | .124 | .148 | .171 | .225 | .281 |
| J- SOCKET WIDTH ACROSS FLATS | MAX | .1111 | .1426 | .1587 | .1900 | .2530 |
| | MIN | .1094 | .1406 | .1562 | .1875 | .2500 |
| E- KEY ENGAGEMENT | MIN | .054 | .077 | .090 | .120 | .151 |
| G- WALL THICKNESS | MIN | .044 | .052 | .061 | .080 | .100 |
| C- RADIUS OR CHAMFER | MAX | .005 | .005 | .005 | .008 | .008 |
| R- RADIUS | MAX | .010 | .012 | .014 | .014 | .017 |
| | MIN | .006 | .007 | .009 | .009 | .012 |
| K- HOLE LOCATION | MAX | .050 | .060 | .065 | .085 | .104 |
| | MIN | .035 | .040 | .045 | .065 | .084 |
| Q- HOLE DIAMETER | MAX | .039 | .050 | .050 | .050 | .050 |
| | MIN | .033 | .044 | .044 | .044 | .044 |
| M- MINIMUM THREAD LENGTH | | .750 | .875 | .875 | 1.000 | 1.125 |
| RECOMMENDED WIRE DIAMETER | MAX | .028 | .040 | .040 | .040 | .040 |
| | MIN | .020 | .032 | .032 | .032 | .032 |
| * DRILL HOLE ALIGNMENT PLUG | | .025 | .030 | .030 | .030 | .030 |
| **MINIMUM BREAKING STRENGTH | | 1,570 LB | 2,890 LB | 3,110 LB | 5,680 LB | 9,100 LB |

| L- LENGTH | PART NO. | PART NO. | PART NO. | PART NO. | PART NO. |
|---|---|---|---|---|---|
| 1/4 | MS 24675 -1 | MS 24675 -6 | MS 24675 -9 | MS 24675-19 | MS24675-30 |
| 3/8 | -2 | -7 | -10 | -20 | -31 |
| 1/2 | -3 | MS 24675 -8 | -11 | -21 | -32 |
| 5/8 | -4 | | -12 | -22 | |
| 3/4 | MS 24675 -5 | | | | -33 |
| 7/8 | | | -13 | -23 | -34 |
| 1 | | | -14 | -24 | -35 |
| 1-1/4 | | | -15 | -25 | -36 |
| 1-1/2 | | | -16 | -26 | -37 |
| 1-3/4 | | | -17 | -27 | |
| 2 | | | MS 24675 -18 | -28 | -38 |
| 2-1/4 | | | | MS24675-29 | -39 |
| 2-1/2 | | | | | MS24675-40 |

(B) FOR CHANGES SEE SHEETS 1, 2, 3

(B) CANCELLED
AFTER 7 JULY 1975 USE MS 24678
SEE SHEET 3

APPROVED 7 APRIL 1960 REVISED (A) 27 JAN 1961 (B) 7 JULY 1975

FSC 5305 CLASS 5305

Project No. 5305-1032

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DD FORM 1 OCT 63 672-1

PROCUREMENT SPECIFICATION FF-S-86

P.A. Navy Other Govt ORD ARD

SUPERSEDES:

TITLE: SCREW, CAP - SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL, UNCOATED (UNF-3A)

SHEET 2

MS 24675

MILITARY STANDARD

| NOMINAL SIZE | | 3/8 | 7/16 | 1/2 | 5/8 |
|---|---|---|---|---|---|
| B- THREADS PER INCH | | 24UNF-3A | 20UNF-3A | 20UNF-3A | 18UNF-3A |
| D- BODY DIAMETER | MAX | .3750 | .4375 | .5000 | .6250 |
| | MIN | .3678 | .4294 | .4919 | .6163 |
| A- HEAD DIAMETER | MAX | .562 | .656 | .750 | .937 |
| | MIN | .550 | .642 | .735 | .921 |
| H- HEAD HEIGHT | MAX | .375 | .437 | .500 | .625 |
| | MIN | .368 | .430 | .492 | .615 |
| F- HOLE DEPTH | MIN | .029 | .031 | .051 | .055 |
| S- HEAD SIDE HEIGHT | MIN | .337 | .394 | .453 | .562 |
| J- SOCKET WIDTH ACROSS FLATS | MAX | .3160 | .3790 | .3790 | .5050 |
| | MIN | .3125 | .3750 | .3750 | .5000 |
| T- KEY ENGAGEMENT | MIN | .182 | .213 | .245 | .307 |
| G- WALL THICKNESS | MIN | .120 | .140 | .160 | .200 |
| C- RADIUS OR CHAMFER | MAX | .008 | .010 | .010 | .010 |
| R- RADIUS | MAX | .020 | .023 | .026 | .032 |
| | MIN | .015 | .018 | .020 | .024 |
| K- HOLE LOCATION | MAX | .123 | .141 | .150 | .198 |
| | MIN | .103 | .121 | .140 | .178 |
| E- HOLE DIAMETER | MAX | .067 | .067 | .067 | .067 |
| | MIN | .061 | .061 | .061 | .061 |
| M- MINIMUM THREAD LENGTH | | 1.250 | 1.375 | 1.500 | 1.750 |
| RECOMMENDED WIRE DIAMETER | MAX | .057 | .057 | .057 | .057 |
| | MIN | .045 | .045 | .045 | .045 |
| ** MINIMUM BREAKING STRENGTH | | 13,800 LB | 18,700 LB | 23,200 LB | 40,400 LB |
| L- LENGTH | | PART NO. | PART NO. | PART NO. | PART NO. |
| 1/2 | | MS 24675 -41 | | | |
| 5/8 | | -42 | | | |
| 3/4 | | -43 | | | |
| 7/8 | | -44 | | | |
| 1 | | -45 | MS 24675 -53 | MS 24675 -57 | MS24675-65 |
| 1-1/4 | | -46 | -54 | -58 | -66 |
| 1-1/2 | | -47 | -55 | -59 | -67 |
| 1-3/4 | | -48 | | -60 | -68 |
| 2 | | -49 | MS 24675-56 | -61 | -69 |
| 2-1/4 | | -50 | | -62 | |
| 2-1/2 | | -51 | | -63 | MS24675-70 |
| 3 | | MS 24675 -52 | | MS 24675 -64 | |

(8) CANCELLED
AFTER 7 JULY 1975 USE MS 24678
SEE SHEET 3

APPROVED 7 APRIL 1960 REVISED (B) FOR CHANGES SEE SHEETS 1, 2, 3

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DD FORM 672-1

P.A. Navy
Other Coast Ord Activities

PROCUREMENT SPECIFICATION
FF-S-86

SUPERSEDING

TITLE
SCREW, CAP - SOCKET HEAD, HEXAGON, DRILLED
ALLOY STEEL, UNCOATED
(UNF-3A)

MILITARY STANDARD

MS 24675

SHEET 3

* Alignment Plug shall pass completely through the head without deflection.

**Based on 160,000 PSI minimum tensile strength. Breaking strengths are determined by the stress areas indicated in Screw-Thread Standards for Federal Services Handbook H-28.

MATERIAL: Steel Alloy; Federal Standard 66

HARDNESS: Rockwell, C 36-43

PROTECTIVE COATING: Uncoated

THREADS: The threads shall be in accordance with Specification MIL-S-7742

THREAD LENGTH-N: Screws which have a body length less than the minimum thread length shall be threaded within two thread pitches of the head.

LENGTH TOLERANCE:

| LENGTH | DIAMETER | TOLERANCE |
|---|---|---|
| 1/4 thru 1.000 | #6 thru 5/8 | ±1/64 |
| 1-1/4 thru 2.000 | #6 thru 5/8 | +1/32 -1/64 |
| 2-1/4 thru 3.000 | #6 thru 5/8 | ±1/32 |

DRILLED HOLE DATA:

(a) Screw size No. 6 shall have two (2) drilled holes spaced 180° and size No. 8 and above shall have six (6) drilled holes spaced 60°
(b) Edges of the holes on the outside of the head shall be chamfered 45° by .010 to .020 deep.
(c) On screw sizes below No. 8 the drilled holes shall be centered as close as practicable on the flat of the hexagon socket.
(d) On screw sizes No. 8 and above the drilled holes shall be within the flats of the hexagon socket and shall not break through the corners of the hexagon socket.

HEAD: Plain or knurled heads are optional.

DIMENSIONS: In inches unless otherwise specified.

NOTES:

(a) This standard takes precedence over documents referenced herein.
(b) Reference documents shall be of the issue in effect on date of invitations for bids.

PART NUMBER = (MS Number) - (Screw dash number)
EXAMPLE = MS24675-1 is the part number for a screw #6-40NF-3A by 1/4 long uncoated steel.

Ⓑ **CANCELLED**

**AFTER 7 JULY 1975 USE MS 24678**

INTERCHANGEABILITY RELATIONSHIP — MS 24678 SCREWS CAN REPLACE THE CANCELLED SCREWS OF THE SAME DASH NUMBERS FOR MOST APPLICATIONS, EXCEPT THOSE IN WHICH CADMIUM PLATED STEEL SCREWS ARE UNACCEPTABLE DUE TO TEMPERATURE, CORROSION, OR TOXICITY CONSIDERATIONS. THE CANCELLED MS 24675 SCREWS CANNOT ALWAYS REPLACE MS 24678 SCREWS AND SHOULD BE USED UNTIL EXISTING STOCK IS DEPLETED. USE ONLY SUPERSEDING MS SCREWS OF THE SAME DASH NUMBERS FOR DESIGN AND REPLACEMENT

FSC 5305

APPROVED 7 APRIL 1960 REVISED Ⓑ FOR CHANGES SEE SHEETS 1, 2, 3

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